

Repeated Queries Exhaust an LLM’s Brand Recommendations but Not Its Sources

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Abstract

Repeating an identical buying question exhausts the set of brands a large language model recommends within about ten runs, while the set of sources it cites is still growing at twenty-four: one run shows 80% of the five-run brand set but only 16–23% of the twenty-four-run source set.

1 Saturation of repeated answers

Auditors, brand trackers, and researchers sample LLM answers to commercial questions and report what they see. Every such report carries an implicit sampling claim: the runs collected have seen enough of what the model would ever say. Prior work established that identical prompts yield different answers run to run [1, 5], and that repetition inside a single prompt changes model behaviour [2]. What has not been measured is the accumulation structure of repeated answers: how fast the set of distinct brands, and the set of distinct cited sources, stops growing as runs accumulate.

We treat runs as sampling units and brands (or cited domains) as species, and apply the classical accumulation toolkit [3, 4]: the exact rarefaction curve $A(k)$, the expected number of distinct items after k of n runs; the final-run yield $A(n) - A(n - 1) = Q_1/n$, where Q_1 counts items seen in exactly one run; and the Chao2 estimate of total repertoire size. All estimators are closed-form; the analysis plan was frozen in version control before any statistic was computed (commit d193c2b).

Figure 1 carries the result. Brand curves flatten: in all four deep cells the 24th run adds nothing. Source curves do not: in all four deep cells the 24th run is still adding domains, with 59–84% of the estimated citation surface seen.

2 Data and results

Breadth: 705 buying-question cells at $n = 5$. We reanalyse our public Category Ownership deposit [7]: 250 brand-free category buying questions (“What is the best CRM for small busi-

Repeated identical questions exhaust the brand list, never the source list

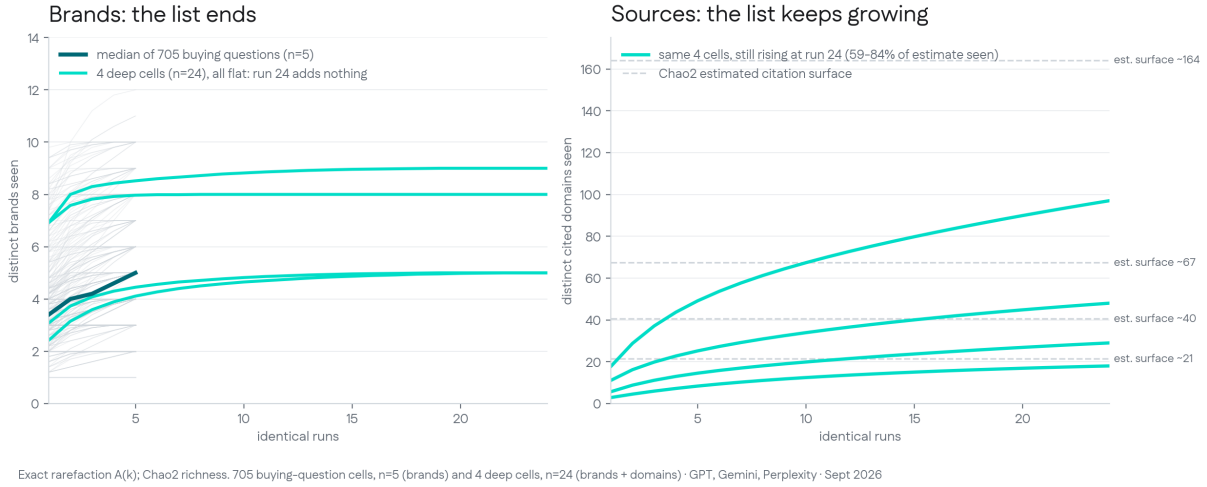


Figure 1: Exact rarefaction curves for distinct brands (left) and distinct cited domains (right) as identical runs accumulate. Left: 705 buying-question cells at $n = 5$ (grey; dark line is the median) and 4 deep cells at $n = 24$ (mint); every deep brand curve is flat by about run ten ($Q_1 = 0$). Right: the same four deep cells on their cited domains; all four are still rising at run 24, against dashed Chao2 estimates of the full citation surface.

nesses?”) across five industries, put to GPT-5.2, Gemini 3 Flash, and Perplexity sonar-pro five times each in June 2026, with brand mentions extracted by the deposit’s own pipeline (3,750 responses). Cells with fewer than four iterations are excluded (32 of 737). Three counting results:

- The repertoire is small and quickly seen. The median cell names 5 distinct brands across five runs, and the median Chao2 share seen at $n = 5$ is 100% (IQR 95–100%). A single run shows 80% of the five-run set (median $A(1)/A(5)$ over 684 complete cells).
- The tail is thin but real. The fifth run still added a brand no earlier run had named in 339 of 705 cells (48%): openai 56%, perplexity 47%, gemini 43% of cells.
- One model is not the market. For one query in four, a single model’s five-run repertoire covers at most 83% of the three-model union (237 queries with all three models).

A case-insensitive dedup of brand strings changes none of these numbers.

Depth: 4 cells at $n = 24$. Two recommendation prompts (UK supermarkets; Nordic care providers), each put 24 times to GPT-5.4 and Gemini 3.5 Flash with web search, June 2026. Brands: all four cells reach $Q_1 = 0$; observed repertoires of 9, 8, 5, and 5 brands are at 95–100% of their final size by run ten, and run 24 adds nothing. Domains: the same four cells have Q_1 of 18, 6, 41, and 12 at run 24; observed surfaces of 48, 18, 97, and 29 domains against Chao2 estimates of 67, 21, 164, and 40 (71%, 84%, 59%, 72% seen); a single run shows 16–23% of the 24-run surface.

The asymmetry is the finding. The brand answer behaves like a small closed vocabulary that repetition exhausts. The citation behaviour behaves like an open population that repetition keeps sampling: at $n = 24$ the Turing estimate Q_1/n still prices the next run at between a quarter and

one-and-three-quarters new domains.

3 Related work

Non-determinism of repeated LLM queries is established [1], and our earlier work decomposes its variance and derives iteration requirements for stable means [5, 6]. This note asks a different question: set growth rather than mean stability. Leviathan et al. [2] show repetition inside one prompt improves accuracy; we measure repetition across calls. Citation instability in generative search is documented on single runs [8, 9]; the accumulation view quantifies how much of the citation surface any fixed sampling budget reveals. The estimators are the species-accumulation classics [3, 4].

4 Conclusion

For anyone measuring AI answers commercially, the two curves price two different products. Brand tracking is bounded: a handful of repeats per question and the answer set is known. Citation monitoring is unbounded at practical budgets: a tracker sampling once per prompt sees roughly a fifth of the surface that twenty-four samples reveal, and the surface has not ended there. What the model says stabilizes; where it looks does not.

Future directions. (1) A breadth replication of the domain curves (the 705-cell corpus stores no citations; this is the one collection the note still wants). (2) Per-language curves on our 16-language repeated-query corpora. (3) Repertoire churn across model versions and time. (4) Good-Turing stopping rules as audit budget policy. (5) Decoding-parameter effects on repertoire size. (6) In-the-wild replication on production audit data. (7) Whether prompt repetition in the sense of [2] shrinks or grows the repertoire.

Declarations

Competing interests. The author is CEO of Rankfor.AI, which sells AI-visibility measurement, including repeated-query brand audits; this note bears on how such measurement should be priced and sampled. **Data and code.** The breadth corpus is our public deposit (Zenodo, 10.5281/zenodo.20788142, CC BY 4.0); the analysis plan, code, per-cell tables, and the frozen-plan commit are at <https://github.com/Rankfor/rankfor-open>. **Provenance.** Secondary analysis of the author’s own deposits; the accumulation analysis appears in no prior manuscript.

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